

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 5: Mercator & Transverse Mercator Projections

Syllabus Source: CAE Oxford (pp. 167-206) | Pass Target: $\geq 90\%$ AviationExam

1. THE DIRECT MERCATOR PROJECTION ARCHITECTURE

The Direct Mercator is a cylindrical conformal projection where the cylinder is tangent to the globe along the Equator:

- Meridians & Parallels: Meridians are parallel, equidistant straight vertical lines. Parallels are straight horizontal lines whose spacing EXPANDS rapidly away from the Equator!
- Rhumb Lines (The Great Benefit): RHUMB LINES ARE STRAIGHT LINES! A straight line drawn between any two points represents a constant compass track. This made Mercator the supreme marine and navigation chart for centuries.
- Great Circles on Mercator: Great Circles are CURVED LINES, bowing towards the nearer pole (lying on the polar side of the rhumb line).
- Scale Variation: Scale is correct at the Equator ($SF = 1.0$), and expands with the secant of latitude: $\text{Scale at Latitude} = \text{Scale at Equator} \times \sec(\text{Latitude}) = \text{Scale at Equator} / \cos(\text{Latitude})$!

2. TRANSVERSE MERCATOR PROJECTION

The cylinder is rotated 90° so it is tangent along a SINGLE MERIDIAN (the Central Meridian):

- Usage: Ideal for North-South flight corridors, topographical national grid surveys (OSGB, UTM), and high-latitude strip charts within 3° of the Central Meridian.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Measuring Distances on a Mercator

Golden Rule of Mercator Charts: - Scale expands continuously towards the poles as $1 / \cos(\text{Latitude})$. - There is NO single graphic distance scale for the entire chart! Mandatory Navigational Procedure: Step 1: Measure the distance between your two waypoints with dividers. Step 2: Place dividers on the LATITUDE scale (vertical border) AT THE MEAN LATITUDE of the route. Step 3: Count minutes of latitude: 1 minute of latitude = 1 NM exact. NEVER use the horizontal longitude scale for distance!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Mercator Chart Scale at High Latitude

SCENARIO (Standard EASA Chart Scale Problem): A Direct Mercator chart has a nominal scale of 1:2,000,000 at the Equator: QUESTION: What is the actual chart scale at 60° North latitude? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Recall the Mercator Scale Expansion Formula: - Scale at Latitude = Scale at Equator x sec(Latitude) - Or in Representative Fraction (RF) denominator terms: - Denominator at Latitude = Denominator at Equator x cos(Latitude)! Step 2: Plug in the numbers for 60° Latitude: - Denominator at Equator = 2,000,000 - Latitude = 60°N → $\cos(60^\circ) = 0.50$ exactly! - Denominator at 60°N = $2,000,000 \times 0.50 = 1,000,000$! Step 3: State the New Scale: - New Scale at 60°N = 1 : 1,000,000! Step 4: Physical Meaning for Dummies: - The scale is twice as large (features appear twice as big on paper) at 60°N than at the Equator, because land masses are stretched by a factor of $1/\cos(60^\circ) = 2.0$! FINAL ANSWER: The scale at 60°N is 1:1,000,000.

[!] EXAM TRAP (AVIATIONEXAM) - The Scale Fraction vs Denominator Trap

- When scale 'EXPANDS' (gets larger), the DENOMINATOR GETS SMALLER! - 1:1,000,000 is a LARGER scale than 1:2,000,000. - Do not multiply the denominator by $\sec(\text{lat})$; multiply the denominator by $\cos(\text{lat})$!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: On a Direct Mercator chart, how do RHUMB LINES appear?**

- [A]: Complex sine curves
- [B]: STRAIGHT LINES
- [C]: Circles centered on the pole
- [D]: Parabolas

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The unique defining property of the Mercator projection is that any straight line drawn across it intersects all meridians at a constant angle, representing a rhumb line.

Q2: How does a GREAT CIRCLE track between two airports in the Northern Hemisphere appear on a Direct Mercator chart?

- [A]: A straight line
- [B]: A CURVED LINE curving towards the North Pole (convex to the Equator)
- [C]: A curved line convex to the pole
- [D]: A vertical line

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Because the Mercator projection stretches higher latitudes, a Great Circle curves northward, bowing away from the Equator towards the pole.

Q3: At what latitude on a Direct Mercator chart is the scale exactly DOUBLE the scale at the Equator?

- [A]: 30°
- [B]: 60° ($\sec(60^\circ) = 1 / \cos(60^\circ) = 1 / 0.50 = 2.0$)
- [C]: 45°
- [D]: 90°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Since scale expands as $\sec(\text{lat})$, at 60° latitude $\sec(60^\circ) = 2.0$, meaning the scale is exactly twice as large as at the Equator.